



Dynamic Evaluation of Decision Criteria in Pharmaceutical Cold Chain Warehousing: A Decision-Analytic Framework under Disruption Risk

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Received: 02-19-2026

Revised: 03-21-2026

Accepted: 03-26-2026

Citation: S. Dabić-Miletić, "Dynamic evaluation of decision criteria in pharmaceutical cold chain warehousing: A decision-analytic framework under disruption risk," *J. Intell. Sustain. Decis. Anal.*, vol. 1, no. 1, pp. 44–57, 2026. <https://doi.org/10.56578/jisda010103>.



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Abstract: Pharmaceutical cold chain warehousing (PCCW) systems operate in highly regulated environments where maintaining product integrity and ensuring continuous operation are critical. In recent years, increasing exposure to systemic disruptions has made it necessary to reconsider how sustainability and resilience criteria are prioritised in warehouse configuration and management. This study aims to investigate how the relative influence of decision criteria evolves under different disruption conditions and to develop a structured analytical framework for evaluating such changes. A decision-analytic framework based on the Decision Criteria Influence (DCI) model was developed. The framework integrated a dual-dimension evaluation of sustainability performance and system reliability with a scenario-based sensitivity adjustment. A structured assessment was conducted across three representative disruption contexts, including energy supply instability, pandemic-induced demand fluctuations, and war-related systemic disruptions. The results showed that under stable conditions, sustainability-oriented criteria, particularly energy efficiency and monitoring-related factors, exerted dominant influence. However, as disruption intensity increased, criteria associated with infrastructure redundancy, inventory buffering capacity, and system reliability became progressively more significant. In extreme scenarios, such as war-related disruptions, resilience-oriented determinants clearly dominated the decision structure, indicating a substantial reordering of strategic priorities. The findings indicate that decision criteria in pharmaceutical cold chain systems exhibit strong context dependency and cannot be treated as static evaluation factors. The proposed framework provides a structured decision-analytic approach for capturing dynamic priority shifts under uncertainty and offers methodological support for designing adaptive and resilient cold chain infrastructures.

Keywords: Pharmaceutical cold chain warehousing; Decision analytics; Multi-criteria evaluation; Disruption risk; Supply chain resilience; Sustainability decision-making

1 Introduction

Pharmaceutical cold chain warehousing (PCCW) systems operate within a highly demanding logistics environment characterized by strict regulatory requirements, technological complexity, and significant public health responsibility [1]. Unlike conventional warehouse facilities, temperature-controlled pharmaceutical storage systems must ensure continuous compliance with quality standards while preserving product integrity throughout storage and handling processes [2]. Failures in such systems extend beyond financial consequences and may directly affect patient safety as well as the continuity of healthcare supply. As a result, PCCW is not only an operational component of logistics infrastructure but also a critical element of healthcare system reliability.

In recent years, the operational context of pharmaceutical logistics has been fundamentally reshaped by large-scale disruptions. Events such as the COVID-19 pandemic, instability in global energy supply, and increasing geopolitical tensions have demonstrated that warehouse configuration and management decisions cannot be based solely on efficiency-oriented considerations [2]. Under conditions of systemic uncertainty, logistics systems are required to maintain operational continuity, recovery capability, and adaptive flexibility. Consequently, criteria related to system robustness, including infrastructure redundancy, inventory buffering capacity, energy security, and operational adaptability, has gained increasing strategic importance in warehouse design and long-term planning.

At the same time, sustainability pressures within pharmaceutical logistics have intensified. Cold warehousing infrastructure represents one of the most energy-intensive segments of logistics operations due to continuous refrigeration requirements, advanced monitoring technologies, and strict environmental control conditions [1]. International institutions, including the World Health Organization and the European Medicines Agency, emphasize the need to ensure both supply reliability and regulatory compliance, while broader policy frameworks increasingly promote energy efficiency and emission reduction within logistics infrastructure [3]. This dual pressure creates a complex decision environment in which operational efficiency must be balanced against environmental performance and system reliability.

Within this context, sustainability-oriented and resilience-oriented criteria represent two interrelated but potentially conflicting dimensions of PCCW system design. Sustainability-oriented criteria, such as energy efficiency, emission reduction, and efficient utilization of storage capacity, focus on minimizing resource consumption and environmental impact [4]. In contrast, resilience-oriented criteria emphasize the capacity of systems to withstand, absorb, and recover from disruptions through measures such as backup infrastructure, safety stock buffers, and spatial diversification of storage capacity [5]. While these measures enhance operational reliability, they may simultaneously increase energy consumption or infrastructure intensity [1, 4]. Conversely, highly optimized and energy-efficient systems may lack the redundancy required to maintain stable operations under extreme disruption conditions. This inherent tension suggests that decision-making in PCCW cannot rely on static or universally prioritised criteria but must account for context-dependent trade-offs between sustainability and resilience.

Despite the growing recognition of this duality, existing research has primarily focused on the operational performance of pharmaceutical supply chains or on isolated aspects of sustainability and resilience [2–6]. Many studies evaluate system outcomes, such as efficiency, cost, or environmental impact, without explicitly analysing the underlying decision criteria that shape these outcomes. Furthermore, limited attention has been devoted to the structural evaluation of decision criteria and to the way in which their relative importance changes under different disruption conditions [7]. In particular, there is a lack of analytical frameworks capable of capturing both the multidimensional nature of decision criteria and their dynamic reconfiguration in response to varying levels of systemic risk. This limitation is critical because warehouse configuration decisions inherently involve multiple interdependent factors, including infrastructure design, automation level, energy systems, monitoring technologies, and spatial organisation of facilities.

Against this background, this study investigates how the relative influence of decision criteria in PCCW systems evolves under different disruption conditions and how these changes affect the prioritisation of sustainability and resilience objectives. The study introduces a structured decision-analytic framework based on the Decision Criteria Influence (DCI) model, designed to evaluate the contribution of individual criteria to both sustainability performance and system reliability. The framework integrates a dual-dimension evaluation approach, capturing the impact of each criterion on sustainability performance and operational continuity, with a scenario-based sensitivity adjustment that reflects changes in disruption intensity.

The proposed framework enables a systematic assessment of how DCI shifts across different operational environments, including stable conditions, energy supply disruptions, pandemic-induced demand fluctuations, and war-related systemic disturbances. By incorporating both structural evaluation and contextual sensitivity, the framework provides a transparent analytical mechanism for examining dynamic changes in decision hierarchies. This approach extends beyond conventional static evaluation methods by explicitly modelling how disruption conditions reshape the relative importance of decision determinants.

The contribution of this study is threefold. First, it develops a structured decision-analytic framework that captures the interaction between sustainability and resilience criteria within PCCW systems. Second, it introduces a scenario-based analytical approach that enables the evaluation of dynamic shifts in decision priorities under different disruption conditions. Third, it provides a transparent modelling structure that supports the systematic comparison of DCI, thereby contributing to the methodological development of sustainability-oriented decision analytics.

From a theoretical perspective, the study advances the understanding of how sustainability and resilience objectives interact within highly regulated and disruption-sensitive logistics systems. From a methodological perspective, it provides a structured analytical tool for evaluating context-dependent decision priorities. From a practical perspective, it offers a basis for designing adaptive warehouse configurations that can respond effectively to changing operational conditions and disruption risks.

2 Analytical Foundations for Decision Criteria Evaluation under Disruption Risk

Decision-making in PCCW cannot be adequately interpreted as a static process governed solely by predefined performance indicators or isolated evaluation metrics. Instead, warehouse configuration decisions emerge from the interaction of multiple decision criteria whose relative importance is contingent upon environmental stability, regulatory constraints, technological conditions, and exposure to systemic disruption risk [1]. Under stable operating conditions, decision frameworks are typically dominated by considerations of economic efficiency and environmental

performance. However, when disruption conditions intensify, the relative importance of criteria shifts toward those associated with operational continuity, recovery capacity, redundancy, adaptability, and energy security [2]. This context dependency suggests that decision criteria must be analysed as dynamically interacting determinants rather than as fixed or universally prioritised variables.

From an analytical perspective, decision criteria in PCCW systems should be conceptualised as interdependent components within a structured evaluation space. Their influence cannot be interpreted independently because the contribution of each criterion is conditioned by its interaction with other criteria and by the broader operational context in which decisions are made [2, 4]. In highly regulated cold chain environments, where temperature integrity, regulatory compliance, and continuous energy supply intersect, decision criteria frequently generate trade-off conditions. For example, investments in backup cooling systems enhance the resilience of cold supply chain operations by reducing recovery time and failure probability, but they simultaneously increase energy consumption and infrastructure complexity [6]. Conversely, highly optimised energy-efficient systems may improve sustainability performance while limiting structural flexibility under emergency conditions. These conflicting effects indicate that decision-making involves balancing competing objectives rather than optimising a single performance dimension.

These interactions can be interpreted through the coexistence of two partially competing strategic logics. The first logic emphasises efficiency-oriented optimisation, focusing on resource minimisation, environmental performance, and cost control [2]. The second logic prioritises robustness and operational continuity, emphasising the capacity of infrastructure to absorb disruptions, maintain service stability, and ensure system recoverability [5]. Disruption risk functions as a moderating factor between these two logics, influencing which criteria assume strategic prominence under specific operating conditions. As disruption intensity increases, the balance between efficiency and robustness shifts, leading to a reconfiguration of decision priorities within PCCW systems.

Within this context, the evaluation of decision criteria requires an analytical framework capable of capturing both their structural interdependencies and their sensitivity to changing operational environments. Rather than assuming a fixed hierarchy of priorities, a decision-analytic perspective must account for the dynamic reweighting of criteria under different disruption scenarios. This implies that the analytical problem is not limited to identifying relevant criteria, but extends to understanding how their relative influence evolves as environmental conditions change. Such a perspective necessitates a modelling structure that integrates both baseline evaluation and context-dependent adjustment mechanisms.

The existing literature provides important insights into various aspects of PCCW, including coordination mechanisms in crisis conditions, resilience modelling, sustainability challenges, and the role of advanced digital technologies. In addition, methodological contributions from multi-criteria decision-making and supply chain risk analysis literature offer relevant foundations for structuring evaluation problems and analysing uncertainty [2–6]. However, most existing studies focus on performance outcomes or on individual dimensions of sustainability and resilience, without explicitly addressing the structural evaluation of decision criteria or their dynamic reconfiguration under disruption conditions [7]. In particular, limited attention has been devoted to the development of analytical frameworks that simultaneously capture (i) the multidimensional nature of decision criteria, (ii) their interdependencies, and (iii) their sensitivity to varying levels of disruption risk. This limitation restricts the ability of existing approaches to support structured decision-making in highly uncertain environments.

To address this gap, it is necessary to establish a conceptual foundation that supports the formulation of a structured decision-analytic model. Such a foundation requires (i) the identification of relevant decision criteria, (ii) the definition of their analytical relationships, and (iii) the specification of mechanisms through which contextual factors influence their relative importance. The analytical perspective adopted in this study treats decision criteria as components of a dynamic evaluation system in which baseline influence is modified by disruption-induced adjustments. This approach provides the conceptual basis for integrating multi-dimensional evaluation with scenario-based sensitivity analysis.

The literature reviewed in this study is synthesised to support the structured identification and classification of decision criteria relevant to PCCW systems. Table 1 presents a consolidated overview of key contributions, highlighting their focus, analytical relevance, and role in informing the proposed evaluation framework. The selected studies collectively provide the theoretical grounding for three essential dimensions of the analytical model: sustainability performance, system resilience, and disruption-driven adaptation. In addition, they inform the methodological choice of structured evaluation and scenario-based adjustment, which underpins the analytical framework developed in the subsequent sections.

Table 1 summarises representative contributions from the literature and illustrates how different research streams contribute to the formulation of the evaluation problem. Studies on cross-sector collaboration and disruption management emphasise the importance of coordination and adaptive response mechanisms [1, 2]. Research on resilience modelling and war-related disruptions highlights the role of infrastructure robustness and systemic vulnerability [3, 5]. Contributions focusing on sustainability challenges and performance evaluation provide the basis for defining efficiency-oriented criteria [6, 8]. Furthermore, methodological studies on multi-criteria decision-

making and supply chain design under uncertainty offer established analytical approaches for structuring evaluation problems and modelling sensitivity to changing conditions [9–11]. Together, these strands of literature support the development of a coherent analytical structure that integrates multiple decision dimensions within a unified evaluation framework.

Table 1. Summary of literature relevant to PCCW

Ref.	Focus of Study	Key Contribution	Relevance to This Study
[1]	Cross sector collaboration in pharmaceutical supply chains	Importance of coordination in crisis management	Supports organizational resilience criteria
[2]	Disruption management in pharmaceutical supply chains	Multi objective optimization under disruptions	Supports scenario based evaluation
[3]	Resilience modeling in pharmaceutical supply chains	Identification of resilience dimensions	Basis for structural resilience criteria
[4]	Blockchain in supply chains	Improves transparency and traceability	Supports monitoring and traceability criteria
[5]	War related pharmaceutical supply disruptions	Impact of conflict on supply availability	Supports extreme disruption scenario
[6]	Sustainability challenges in cold chain logistics	Environmental and operational constraints	Supports sustainability criteria
[7]	Sustainable pharmaceutical cold chains	Performance evaluation of cold chains	Links sustainability and efficiency
[8]	Scalability of pharmaceutical supply chains	Tradeoffs between resilience and sustainability	Supports dynamic importance of criteria
[9]	Multi criteria decision making methods	Structured evaluation of multiple criteria	Methodological basis for scoring approach
[10]	Supply chain design under uncertainty	Modeling uncertainty and risk	Supports scenario sensitivity concept
[11]	Ripple effect in supply chains	Propagation of disruptions	Supports inclusion of disruption scenarios

Note: PCCW = pharmaceutical cold chain warehousing.

Building upon these conceptual and methodological foundations, the following section establishes a structured classification of decision criteria relevant to PCCW systems. This classification serves as the analytical input for the decision-analytic model and provides the basis for evaluating how criteria influence varies under different disruption scenarios. By linking conceptual understanding with analytical structuring, this study ensures that the proposed modelling framework is both theoretically grounded and methodologically consistent.

3 Structured Classification of Decision Criteria in Pharmaceutical Cold Chain Warehousing

This section identifies and organizes the key decision criteria relevant to PCCW. The proposed structure is based on the premise that decision criteria are context sensitive and mutually interconnected. Rather than treating sustainability and resilience as separate dimensions, the framework groups the identified criteria into three analytically related categories: operational sustainability, structural resilience, and enabling system capacity [8]. This classification reflects the reality that pharmaceutical cold warehouses operate simultaneously as energy-intensive facilities, risk-sensitive infrastructures, and digitally supported systems. As a result, warehouse configuration decisions must consider environmental performance, operational reliability, and the technological and organizational capabilities that support the coordination of these objectives [5]. Operational sustainability criteria refer to factors that influence energy efficiency and resource utilization within cold storage operations. Structural resilience criteria capture the capacity of warehouse infrastructure to maintain stable operations and recover from disruptions [7]. Enabling system capacity includes digital technologies and organizational capabilities that support monitoring, coordination, and adaptive responses to changing operational conditions. By structuring decision criteria into these three categories, the framework provides a clearer analytical basis for evaluating how different determinants influence warehouse configuration decisions under both stable and disrupted operating environments. This structured classification serves as the analytical input for the decision-analytic model and provides the basis for evaluating criteria influence in subsequent sections.

3.1 Operational Sustainability Criteria

Operational sustainability criteria represent key determinants in evaluating environmental performance and resource efficiency within the analytical framework. Those criteria refer to factors that determine the environmental performance and resource efficiency of cold warehousing operations. These criteria capture the ability of storage infrastructure to minimize energy consumption, reduce emissions, and ensure efficient utilization of physical and technological resources while maintaining strict temperature control requirements. These criteria are typically dominant under stable market conditions and regulatory emphasis on decarbonization. Key elements include [2–7]:

1. Energy intensity of refrigeration systems
2. Carbon emission profile of warehouse operations
3. Efficiency of temperature control mechanisms
4. Resource utilization rates (space, equipment, packaging)
5. Waste minimization and product loss prevention
6. Integration of renewable or low-carbon energy sources

In PCCW, energy consumption represents one of the main sustainability drivers. Decisions related to insulation quality, cooling technology, automation levels, and energy sourcing directly influence long-term environmental performance.

3.2 Structural Resilience Criteria

Structural resilience criteria represent the capacity of warehouse infrastructure to maintain stable operations and recover from operational disturbances. These criteria reflect the presence of redundancy, backup systems, and buffering mechanisms that enable pharmaceutical cold storage facilities to preserve product integrity during disruptions. Relevant criteria include [2–6]:

1. Redundancy of cooling and power systems
2. Backup energy availability
3. Inventory buffering capacity
4. Network decentralization or facility diversification
5. Recovery time after operational interruption
6. Adaptability to demand surges

In cold warehousing, even short-term power outages may compromise entire inventory batches. Therefore, redundancy and energy security assume strategic importance. However, these measures often increase infrastructure intensity and resource consumption, reinforcing the sustainability–resilience tension.

3.3 Enabling System Criteria

Enabling system criteria refer to technological and organizational capabilities that support both sustainability and operational reliability. These determinants enhance monitoring, coordination, and adaptive responses within cold chain systems through digital technologies, automation, and data driven decision support. They include [5, 6, 8]:

1. Real-time temperature monitoring systems
2. Digital traceability and visibility platforms
3. Predictive maintenance capabilities
4. Automation and robotics integration
5. Data-driven demand forecasting
6. Organizational response protocols and training

Digital technologies, such as IoT-based monitoring and predictive analytics, can reduce energy waste while simultaneously enhancing early disruption detection. In this sense enabling system capacity functions as a bridging layer between efficiency and robustness objectives [9, 10]. The criteria identified above form the analytical basis for the evaluation framework applied in the following section. Their relative influence is assessed in order to examine how decision priorities in PCCW may shift under different disruption conditions.

4 Methodological Framework for Evaluating Decision Criteria Influence

This study develops a structured decision-analytic framework to evaluate the relative influence of decision criteria in PCCW systems under varying disruption conditions. The framework is designed to capture both the intrinsic contribution of individual criteria and the dynamic adjustments in their importance induced by changes in the operational environment.

The analytical structure integrates a dual-dimension evaluation of decision criteria with a scenario-based sensitivity adjustment mechanism. Each criterion is assessed in terms of its contribution to sustainability performance and system reliability, reflecting the two fundamental dimensions of decision-making in PCCW systems. To represent the baseline structural importance of each criterion, a composite influence measure is defined. This baseline

assessment is subsequently adjusted through a contextual sensitivity factor that captures the impact of disruption intensity on decision priorities.

By combining baseline influence evaluation with scenario-dependent adjustment, the framework enables a systematic examination of how decision hierarchies evolve under different disruption conditions. This approach allows the identification of shifts in strategic priorities and provides a structured basis for comparing the relative importance of decision criteria across multiple operational scenarios. The resulting methodological structure supports a transparent and analytically consistent evaluation of decision-making dynamics in disruption-sensitive logistics systems.

4.1 Baseline Influence and Scenario Sensitivity Assessment

In the first phase of the analysis, each identified decision criterion is evaluated across two independent dimensions:

Impact on sustainability performance (IS)

Impact on system reliability and operational continuity (IR)

Both dimensions are assessed using a five point Likert scale, a common approach in multi criteria evaluation and decision analysis studies [5, 6]:

1 = very low influence

2 = low influence

3 = moderate influence

4 = strong influence

5 = very strong influence

The assigned values are based on insights from existing literature on cold chain logistics, sustainability in warehousing, and supply chain resilience, as well as on general operational characteristics of pharmaceutical cold storage systems [3]. Those values should be interpreted as representative and theoretically grounded estimates used for analytical illustration, rather than empirically validated measurements [7, 9]. This dual evaluation allows each criterion to be positioned within a two-dimensional analytical space that captures both environmental and operational contributions analytical space that captures both environmental and operational contributions, consistent with multi-criteria decision making approaches [9, 10]. Similar scoring based approaches are frequently applied in logistics and supply chain decision models to quantify the relative influence of operational determinants [3, 11]:

$$BI = IS + IR$$

where, BI represents the Base Influence score. This additive aggregation is commonly applied in scoring-based multi-criteria evaluation models to represent the overall influence of decision factors [2, 9]. To reflect the changing nature of operational environments in PCCW, which is consistent with scenario-based analysis used in supply chain resilience studies [1]:

1. Induced demand surge, characterized by sudden increases in pharmaceutical demand and operational pressure on storage and distribution capacity;

2. Energy supply disruption, involving electricity price volatility, power shortages, or instability affecting refrigeration systems;

3. Regulatory tightening, reflecting stricter temperature control, traceability requirements, and environmental reporting obligations.

Changes in the importance of decision criteria under these conditions are represented through a Scenario Sensitivity Factor (SSF) representing simplified sensitivity adjustments frequently applied in scenario modeling and decision support frameworks [3, 8]:

0.8 = decreased importance

1.0 = unchanged importance

1.2 = moderately increased importance

1.5 = strongly increased importance

The SSF is introduced as a simplified analytical adjustment proposed in this study, conceptually grounded in scenario-based sensitivity analysis approaches commonly used in supply chain and logistics research [8, 11]. The asymmetric scale is intentionally applied to reflect the assumption that disruption conditions are more likely to amplify the importance of certain criteria than to significantly reduce it, while maintaining model simplicity and interpretability.

4.2 Decision Criteria Influence Index

The final influence value of each decision criterion under a given disruption condition is calculated using the DCI:

$$DCI = BI \times SSF$$

Given the structure of the model, DCI values are bounded within a finite range determined by the Likert scale and SSFs. For improved comparability, DCI values may also be normalized relative to the maximum value within each scenario.

The introduction of disruption scenarios reflects the growing body of research emphasizing supply chain resilience and the need to evaluate system behavior under uncertain operating conditions [11]. The purpose of the DCI framework is not to identify an optimal warehouse configuration, but to illustrate possible shifts in decision priorities when the operational environment changes. Table 2 presents an example of the operationalization of the DCI framework. The BI score reflects the intrinsic contribution of each criterion, while the SSF represents contextual adjustments under disruption conditions. The resulting DCI values indicate the adjusted importance of individual criteria within the analyzed environment.

Table 2. DCI assessment under energy supply disruption

Criterion	IS	IR	BI	SSF	Scenario	DCI
Energy efficient cooling systems	5	3	8	1.2	Energy crisis	9.6
Backup power redundancy	2	5	7	1.5	Energy crisis	10.5
Real time temperature monitoring	4	4	8	1.3	Energy crisis	10.4

Note: DCI = Decision Criteria Influence; BI = Base influence core; IS = Impact on sustainability performance; IR = Impact on system reliability and operational continuity; SSF = Scenario Sensitivity Factor.

The table illustrates how disruption conditions act as contextual modifiers that reshape managerial priorities in PCCW systems. By incorporating the SSF into the analytical structure, the framework captures the dynamic reconfiguration of decision hierarchies. This approach highlights the role of disruption risk as a moderating force that can shift the relative dominance of different decision criteria within strategic warehouse configuration processes. The analytical structure presented above establishes the methodological basis for evaluating the influence of decision criteria under different operational conditions. In order to examine the implications of this framework, the following section applies the proposed approach to the selected disruption contexts and evaluates the resulting changes in the relative importance of decision criteria in PCCW systems.

To improve the comparability of decision criteria across different disruption scenarios, the DCI values can be normalized relative to the maximum value observed within a given scenario ($DCI_{norm} = DCI/DCI_{max}$). This normalization transforms the results into a standardized scale ranging from 0 to 1, allowing for a clearer interpretation of the relative importance of each criterion. Table 3 presents an illustrative example of normalized DCI values under the energy supply disruption scenario.

Table 3. Example of Normalized DCI Values under energy supply disruption

Criterion	BI	SSF	DCI	DCI_norm
Energy efficiency of refrigeration systems	9	1.2	10.8	0.72
Temperature control precision	10	1.5	15.0	1.00
Backup power redundancy	8	1.5	12.0	0.80
Inventory buffering capacity	7	1.2	8.4	0.56
Waste and product loss prevention	6	1.0	6.0	0.40

Note: DCI = Decision Criteria Influence; BI = Base Influence score; SSF = Scenario Sensitivity Factor.

The normalized values highlight the relative dominance of specific criteria, making it easier to compare their influence within the same analytical context.

5 Analytical Framework for Evaluating Decision Criteria Influence

This section presents the analytical procedure used to examine the influence of selected decision criteria in PCCW systems. The proposed Decision Criteria Influence framework is applied to evaluate the relative importance of criteria associated with sustainability and system reliability across different disruption environments. The analysis considers several types of disturbances that are relevant for pharmaceutical cold chain operations, including energy supply instability, pandemic-driven demand fluctuations, and war induced systemic disturbances. Evaluating the influence of decision criteria under these conditions allows the identification of potential shifts in decision priorities and changes in the relative importance of key determinants affecting warehouse configuration.

5.1 Design of the Analytical Evaluation Framework

To demonstrate the analytical applicability of the proposed DCI framework, this study adopts a structured scenario-based analytical approach. Instead of relying on empirical survey data, representative baseline values are

assigned to each decision criterion based on theoretically grounded assumptions derived from sustainability and resilience research in cold chain logistics [12, 13]. The objective of this analytical evaluation is to illustrate how different disruption scenarios may modify the relative importance of sustainability and resilience-oriented decision criteria. The assigned numerical values, consequently, reflect plausible influence patterns under stable operating conditions as well as under selected disruption environments. This analytical design enables conceptual clarity while maintaining methodological transparency. By examining potential changes in decision weights across different disruption scenarios, the framework provides a structured perspective on the interaction between sustainability and resilience determinants, highlighting shifts in priority structures that may be difficult to capture through static evaluation approaches.

Scenario-adjusted DCI values were derived by integrating baseline criterion importance with SSF, reflecting the varying impact of different disruption contexts on operational priorities. The resulting values represent scenario-conditioned importance indices used for comparative analysis across disruption types, rather than strictly linear transformations of baseline scores.

5.2 Baseline Influence Assessment under Stable Conditions

In the baseline scenario representing stable operating conditions, each decision criterion is evaluated along two independent dimensions: its impact on IS and its impact on IR. Both dimensions are measured using a five-point scale, where higher values indicate a stronger influence. The BI score is calculated as the sum of IS and IR values. This aggregation captures the intrinsic structural importance of each criterion within PCCW systems when disruption risk remains moderate.

Under stable conditions, sustainability-driven determinants such as energy efficiency, carbon performance, and digital monitoring are assumed to exert a strong influence due to regulatory pressure and cost optimization priorities [2]. Resilience-related determinants, including redundancy and buffering capacity, remain relevant but are not dominant in the absence of acute disruption [6, 14].

The baseline influence values presented in Table 4 reflect the assumed structural importance of each criterion under stable operating conditions. Sustainability-oriented criteria, such as energy efficiency and carbon performance, receive high IS scores, while resilience-oriented determinants such as backup power and system redundancy exhibit stronger IR contributions. Digital and monitoring-related criteria demonstrate balanced influence across both dimensions, positioning them as integrative enablers within PCCW systems [4–8].

Table 4. Analytical baseline influence assessment under stable operating conditions

Criterion	Decision Criterion	IS	IR	BI (= IS + IR)
C1	Energy efficiency of refrigeration systems	5	3	8
C2	Carbon emission intensity	5	2	7
C3	Temperature control precision	4	4	8
C4	Resource utilization efficiency	4	3	7
C5	Waste and product loss prevention	4	3	7
C6	Backup power redundancy	2	5	7
C7	Cooling system redundancy	2	5	7
C8	Inventory buffering capacity	3	4	7
C9	Facility diversification	3	4	7
C10	Real-time temperature monitoring	4	4	8
C11	Predictive maintenance capability	4	4	8
C12	Automation and workforce flexibility	3	4	7

Note: BI = Base Influence score; IS = Impact on sustainability performance; IR = Impact on system reliability and operational continuity.

5.3 Decision Criteria Influence under Energy Supply Instability

Energy supply instability represents a critical operational risk for PCCW systems due to their continuous dependence on reliable refrigeration and environmental control technologies [1]. Even short interruptions in electricity supply may compromise temperature sensitive pharmaceutical products and threaten regulatory compliance [7]. In addition to supply interruptions, sharp increases in energy costs may also influence the operational stability of cold storage infrastructure. Under such conditions, decision criteria associated with infrastructure reliability and energy continuity become particularly important. Backup power systems, cooling system redundancy, and predictive maintenance capabilities play a key role in ensuring uninterrupted operation and preventing temperature deviations. These criteria support the ability of cold warehousing systems to maintain product integrity even under unstable energy supply conditions. Within the analytical framework applied in this study, the relative influence of selected

decision criteria was evaluated under the energy disruption scenario. The adjusted DCI values presented in Table 5 illustrate the increased importance of criteria associated with infrastructure reliability and technical redundancy compared with the baseline configuration [6–11, 15].

Table 5. Scenario-adjusted DCI results under energy supply disruption

Criterion	Decision Criterion	SSF (Energy Crisis)	DCI
C1	Energy efficiency of refrigeration systems	1.3	10.4
C2	Carbon emission intensity	1.0	7.0
C3	Temperature control precision	1.2	9.6
C4	Resource utilization efficiency	1.0	7.0
C5	Waste and product loss prevention	1.0	7.0
C6	Backup power redundancy	1.5	10.5
C7	Cooling system redundancy	1.4	9.8
C8	Inventory buffering capacity	1.1	7.7
C9	Facility diversification	1.2	8.4
C10	Real-time temperature monitoring	1.2	9.6
C11	Predictive maintenance capability	1.3	10.4
C12	Automation and workforce flexibility	1.1	7.7

Note: DCI = Decision Criteria Influence; SSF = Scenario Sensitivity Factor.

5.4 Decision Criteria Influence under Pandemic Disruption

Pandemic disruptions create a specific operational environment for PCCW systems [1, 9]. Unlike energy instability, which primarily affects infrastructure reliability, pandemic conditions typically generate sudden fluctuations in demand, increased throughput requirements, and operational constraints related to workforce availability and mobility restrictions. During global health emergencies, supply chains for temperature-sensitive pharmaceutical products experience significant pressure due to increased demand for vaccines, biological medicines, and diagnostic materials [13, 14]. Under such circumstances, warehouse systems must demonstrate a higher level of operational flexibility and responsiveness. Decision criteria associated with inventory buffering capacity, system scalability, and digital monitoring become particularly relevant, as they support the ability of warehouses to accommodate demand volatility while maintaining strict temperature control and regulatory compliance [1, 4, 16]. Automated handling technologies and digital monitoring systems also contribute to operational stability by reducing dependence on manual processes and improving system visibility. Within the analytical framework applied in this study, the relative influence of selected decision criteria was evaluated under pandemic disruption conditions. The results presented in Table 6 illustrate the adjusted DCI values that reflect the increased importance of criteria associated with operational flexibility, inventory adaptability, and digital monitoring capability in comparison with the baseline configuration [1–4, 9].

Table 6. Scenario-adjusted DCI results under pandemic disruption

Criterion	Description	Pandemic Scenario Multiplier	DCI
C1	Energy efficiency of refrigeration systems	1.0	8.0
C2	Carbon emission intensity	0.9	5.4
C3	Temperature control precision	1.0	8.0
C4	Resource utilization efficiency	1.0	6.0
C5	Waste and product loss prevention	1.2	8.4
C6	Backup power redundancy	1.1	7.7
C7	Cooling system redundancy	1.1	7.7
C8	Inventory buffering capacity	1.5	9.0
C9	Facility diversification	1.2	7.2
C10	Real time temperature monitoring	1.4	11.2
C11	Predictive maintenance capability	1.2	9.6
C12	Automation and workforce flexibility	1.3	9.1

Note: DCI = Decision Criteria Influence.

5.5 War-Induced Systemic Disruption Scenario

Disruptions caused by wars represent the most complex disturbance environment for PCCW systems. In addition to supply chain interruptions, such conditions may involve infrastructure damage, transportation instability, energy insecurity, and heightened cyber and regulatory risks [3]. These disruptions create a highly uncertain operational environment that requires multiple layers of system resilience. Under such circumstances, decision criteria related to infrastructure redundancy, spatial diversification of facilities, and digital system reliability become critically important [8]. Redundant cooling systems, geographically distributed storage capacity, and robust digital monitoring infrastructure help ensure operational continuity even when parts of the logistics network are disrupted [13, 16]. These measures strengthen the ability of PCCW systems to maintain supply reliability in highly unstable environments. Within the analytical framework applied in this study, the influence of the selected decision criteria was evaluated under war-related disruption conditions. The DCI values presented in Table 7 illustrate the strong prioritization of resilience-oriented determinants that support infrastructure robustness, spatial flexibility, and digital system stability [1–5, 17].

Table 7. Scenario-adjusted DCI results under war-related disruption

Criterion	Criterion	SSF (War)	DCI
C1	Energy efficiency of refrigeration systems	1.2	9.6
C2	Carbon emissions intensity	0.9	6.3
C3	Temperature control precision	1.3	10.4
C4	Resource utilization efficiency	0.9	6.3
C5	Waste and product loss prevention	1.1	7.7
C6	Backup power redundancy	1.6	11.2
C7	Cooling system redundancy	1.5	10.5
C8	Inventory buffering capacity	1.5	10.5
C9	Facility diversification	1.6	11.2
C10	Real-time temperature monitoring	1.4	11.2
C11	Predictive maintenance capability	1.4	11.2
C12	Automation and workforce flexibility	1.3	9.1

Note: DCI = Decision Criteria Influence; SSF = Scenario Sensitivity Factor.

The war-induced disruption scenario produces the most pronounced reordering of decision priorities among all analyzed environments. Backup power redundancy (C6), facility diversification (C9), real-time monitoring (C10), and predictive maintenance (C11) reach the highest DCI values, indicating the critical importance of infrastructural autonomy and system visibility under extreme instability. Inventory buffering (C8) also demonstrates substantial amplification, reflecting the necessity of stockpiling strategies when transportation routes are uncertain [3–12]. In contrast, carbon emission intensity (C2) and resource utilization efficiency (C4) experience relative decline, suggesting that immediate operational survival temporarily outweighs long-term sustainability optimization.

Compared to the energy crisis scenario, the war scenario introduces broader systemic amplification across both physical and digital resilience determinants. This indicates that multidimensional disruptions simultaneously elevate infrastructure redundancy, geographic decentralization, and cyber-physical system integrity. The analytical framework thus suggests that PCCW systems under wartime conditions transition from efficiency-oriented optimization toward survival-driven strategic structuring.

5.6 Comparative Assessment of Decision Criteria Influence across Disruption Scenarios

To examine the impact of disruption intensity on decision priorities in PCCW systems, a comparative scenario analysis was conducted. The DCI values were evaluated simultaneously to identify potential shifts in the relative importance of sustainability and resilience-related determinants. This comparative perspective provides a clearer interpretation of the ways in which different categories of disruption reshape the decision environment and alter the hierarchy of criteria relevant for warehouse configuration. Table 8 presents a comparative overview of DCI values across three disruption scenarios: energy crisis, pandemic-induced demand surge, and war-induced systemic instability. The results indicate a progressive amplification of resilience-oriented determinants as the severity and multidimensional nature of disruption intensify.

The comparison between baseline influence values and scenario adjusted DCI scores indicates a substantial restructuring of decision priorities once disruption conditions are introduced. Under stable operating conditions, the highest baseline influence values are associated with energy efficiency (C1), temperature control precision (C3), real time monitoring (C10), and predictive maintenance capability (C11) [3–7, 18]. These criteria reflect the dominant importance of operational sustainability and digital control mechanisms in routine PCCW environments, where

system performance is primarily evaluated through efficiency, environmental performance, and quality assurance indicators.

Table 8. Scenario-adjusted DCI results under war-related disruption

Criterion	Criterion	Energy Crisis DCI	Pandemic DCI	War DCI
C1	Energy efficiency of refrigeration systems	10.4	8.8	9.6
C2	Carbon emissions intensity	7.0	7.0	6.3
C3	Temperature control precision	9.6	8.8	10.4
C4	Resource utilization efficiency	7.0	7.0	6.3
C5	Waste and product loss prevention	7.0	7.7	7.7
C6	Backup power redundancy	10.5	8.4	11.2
C7	Cooling system redundancy	9.8	8.4	10.5
C8	Inventory buffering capacity	7.7	10.5	10.5
C9	Facility diversification	8.4	7.7	11.2
C10	Real-time temperature monitoring	9.6	10.4	11.2
C11	Predictive maintenance capability	10.4	9.6	11.2
C12	Automation and workforce flexibility	7.7	9.8	9.1

Note: DCI = Decision Criteria Influence.

The introduction of disruption scenarios significantly modifies this priority structure. In the energy crisis scenario, criteria related to infrastructure reliability and energy continuity experience the strongest increase in influence. Backup power redundancy (C6) rises sharply and becomes the most influential determinant, while cooling system redundancy (C7) and predictive maintenance capability (C11) also gain substantial importance [1–4, 8, 19]. These shifts reflect the critical dependence of PCCW on uninterrupted power supply, where even short-term energy interruptions may jeopardize temperature sensitive pharmaceutical inventories [4–10, 20]. In such conditions, resilience oriented determinants that secure operational continuity become strategically dominant. In contrast, sustainability oriented criteria such as carbon emission intensity (C2) and resource utilization efficiency (C4) maintain relatively stable influence levels and do not exhibit comparable amplification.

The pandemic disruption scenario produces a different adjustment of priorities. In this environment, the relative importance of operational flexibility and system responsiveness becomes more pronounced. Inventory buffering capacity (C8), real time monitoring (C10), and automation capability (C12) gain stronger influence, reflecting the need for rapid throughput adjustments, demand variability management, and reduced dependence on manual operations during workforce instability [11–19]. These criteria support the ability of warehouse systems to absorb sudden demand surges while maintaining product integrity and regulatory compliance.

The war induced disruption scenario generates the most extensive reconfiguration of decision priorities. Under conditions of systemic instability, infrastructure redundancy (C6 and C7), facility diversification (C9), and digital system reliability (C10 and C11) reach the highest influence values across all examined scenarios [1, 3–7]. This pattern indicates that extreme disruption environments simultaneously elevate multiple layers of resilience, including physical infrastructure robustness, spatial diversification of storage facilities, and cyber physical system stability. At the same time, criteria primarily associated with long term environmental optimization, including carbon emission performance (C2), exhibit a relative decline in strategic importance. This shift reflects a temporary reorientation of priorities from sustainability optimization toward operational survivability and supply continuity [13, 20].

Figure 1 illustrates the variation of Scenario-Adjusted DCI values across disruption conditions. The results indicate a clear shift in decision priorities under war-related disruption, where resilience-oriented criteria dominate over sustainability-oriented factors, highlighting the increasing importance of system robustness and operational continuity in highly unstable environments.

Figure 1 suggests that the highest DCI values under the war-related disruption scenario are associated with C6 (Backup power redundancy), C7 (Cooling system redundancy), and C9 (Facility diversification). These criteria consistently indicate peak influence levels, highlighting their critical role in ensuring operational continuity and maintaining temperature integrity under conditions of severe instability. Their dominance reflects the increased importance of redundancy and spatial flexibility when infrastructure reliability is uncertain. In contrast, criteria such as C2 (Carbon emission intensity) and C5 (Waste and product loss prevention) display comparatively lower DCI values, indicating a reduced relative importance of sustainability-oriented objectives under extreme disruption conditions. This divergence confirms a clear shift in decision priorities, where resilience-related determinants outweigh efficiency and environmental considerations in high-risk environments.

The comparative analysis confirms that disruption severity functions as a moderating factor that dynamically reshapes the balance between sustainability and resilience oriented decision criteria. While sustainability determinants dominate the evaluation structure under stable conditions, resilience capacity gradually supersedes efficiency oriented

objectives as systemic risk intensifies. An important observation emerging from all scenarios concerns the consistent influence of digital enabling criteria, particularly real time monitoring and predictive maintenance capabilities. These determinants retain strong strategic importance across different disruption environments, suggesting their integrative role in supporting both sustainability and resilience objectives within PCCW systems.

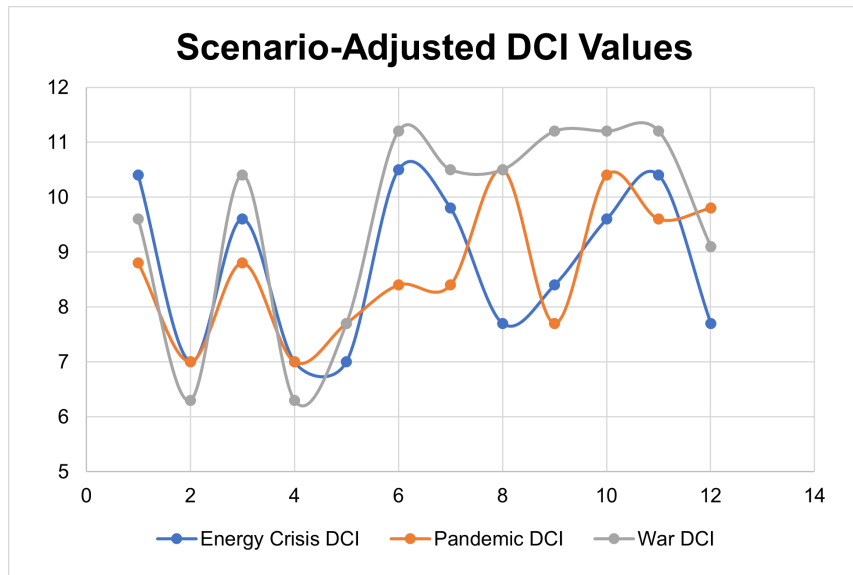


Figure 1. Scenario-adjusted DCI results under disruption conditions

Note: DCI = Decision Criteria Influence

6 Conclusions

This study developed a structured decision-analytic framework to evaluate the influence of sustainability and resilience criteria in PCCW systems under varying disruption conditions. By introducing the DCI model within a scenario-based analytical structure, the study demonstrates how disruption contexts reshape decision hierarchies and alter strategic priorities in warehouse configuration and management.

The results reveal that decision criteria in PCCW systems exhibit strong context dependency. Under stable operating conditions, sustainability-oriented determinants, particularly energy efficiency and digital monitoring, dominate the evaluation structure. As disruption intensity increases, resilience-oriented criteria progressively gain prominence. Infrastructure redundancy and system reliability become critical under energy supply instability, while operational flexibility and buffering capacity emerge as key determinants under pandemic conditions. In extreme disruption environments, such as war-related instability, multilayer resilience mechanisms—including energy autonomy, spatial diversification, and digital system robustness—become the dominant drivers of decision-making.

These findings demonstrate that sustainability and resilience should not be interpreted as competing objectives with fixed trade-offs. Instead, their relative importance is dynamically reconfigured depending on disruption severity. In this respect, the proposed framework contributes to the literature by establishing a transparent analytical mechanism for modelling the dynamic reordering of decision criteria under uncertainty. It provides a structured approach for integrating multi-dimensional evaluation with scenario-based sensitivity, thereby advancing decision-analytic modelling in sustainability-oriented logistics systems.

From a practical perspective, the results highlight the need for adaptive decision frameworks in PCCW. Approaches that rely solely on efficiency-oriented optimisation may be insufficient in disruption-prone environments. Instead, warehouse design and investment decisions should incorporate scenario-based evaluation and stress-testing mechanisms that account for potential shifts in critical determinants. In addition, the consistent influence of digital enabling criteria, such as real-time monitoring and predictive maintenance, underscores the integrative role of digitalisation in balancing sustainability and resilience objectives.

This study is subject to several limitations. First, the proposed framework is conceptual and relies on theoretically grounded assumptions rather than empirical data. Second, the baseline influence scores and sensitivity factors are illustrative and may not fully capture real-world variability. Third, the analysis is limited to a predefined set of disruption scenarios. Future research may extend this work through empirical validation, expert-based evaluation, and the inclusion of additional disruption types, such as cyber risks or regulatory shocks. Such extensions would further enhance the robustness and applicability of the proposed framework.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The author declares that generative AI tools were used solely for language editing and improving the readability of the manuscript. All scientific content, analysis, and interpretations were developed and verified by the authors, who take full responsibility for the integrity of the work.

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